

Soumyadeep Nath

Aspiring Data Scientist & AI Builder | Python · SQL · ML · GenAI · Full-Stack AI

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PROFESSIONAL SUMMARY

From literature to data — building the bridge deliberately. Currently completing an Executive PG Programme in Data Science & AI at IIT Roorkee, working hands-on with Python, SQL, Machine Learning, and Generative AI systems including RAG pipelines, embeddings, and LLM APIs. Built and deployed two live AI applications: **AlignResume**, a production-grade full-stack AI platform with truthfulness guardrails and structured JSON prompt pipelines, and **Future Fit**, a Streamlit-based skill trend analytics platform built on real job market data. A background in academic writing and language analysis provides a unique edge: translating complex model outputs into clear, compelling narratives that stakeholders understand.

EDUCATION & TRAINING

Executive PG Programme in Data Science & AI – In Progress

Feb 2026 – Present

Indian Institute of Technology (IIT) Roorkee | Remote

Coursework: Python · SQL · NumPy · Pandas · Machine Learning · Statistics · Data Visualization · Generative AI · Prompt Engineering · RAG · LLM APIs · Embeddings · Vector Databases

- Engineered data pipelines and automated ETL workflows using Python (Pandas, NumPy) to clean, transform, and prepare structured and unstructured datasets for downstream analysis.
- Queried relational databases using advanced SQL — window functions, joins, subqueries, and aggregations — extracting actionable, analysis-ready business insights.
- Constructed EDA frameworks applying statistical reasoning, hypothesis testing, and A/B testing methodology to validate data-driven decisions.
- Developed and evaluated supervised Machine Learning models — regression, classification, and tree-based ensembles (Random Forest, Gradient Boosting) — using scikit-learn.
- Implemented Generative AI applications by integrating LLM APIs, engineering structured prompts, and building RAG systems with embeddings and FAISS/Chroma vector databases.
- Translated complex model outputs into structured data narratives and interactive dashboards using Matplotlib, Seaborn, and Plotly; documented findings in reproducible Jupyter notebooks.

M.A. in English Literature — First Division

Oct 2022 – Jul 2024

Presidency University, Kolkata

B.A. Honours in English Literature — First Division

Sep 2019 – Jul 2022

Presidency University, Kolkata

PROJECTS

AlignResume — AI-Powered Resume Optimization Platform

May 2026

Live: align-resume-beta.vercel.app · GitHub: github.com/sdn9300/align-resume · Stack: Next.js · TypeScript · Tailwind CSS · Groq API (LLaMA 3.3 70B) · Playwright · Zod · Vercel

- Architected and deployed an end-to-end full-stack AI web application that parses resumes and job descriptions, scores alignment quality, and generates ATS-optimized bullet rewrites via a structured JSON prompt pipeline over the Groq API — with Zod schema validation ensuring deterministic, parseable LLM outputs.
- Implemented an automated truthfulness guardrail layer that validates rewritten bullets against the original resume, blocking export if hallucinated metrics, employers, or skills are detected — prioritizing honesty over inflation.
- Engineered Playwright-based headless PDF export, rate-limited API proxy (10 runs/hr per IP), and GitHub-Vercel CI/CD auto-deploy; deepened expertise in LLM application architecture, prompt engineering, and production-grade full-stack AI deployment.

Future Fit — AI-Powered Skill Trend Analysis Platform

Jun 2026

Live: skilltrendanalysis-rf3zefgsjaa4l9f8pu2prb.streamlit.app · GitHub: github.com/sdn9300/Future-Fit-AI-Powered-Skill-Trend-Analysis · Stack: Python · Streamlit · Pandas · Plotly

- Developed an interactive analytics platform that processes job market data to surface emerging skill trends and in-demand technologies — enabling data-driven career planning and workforce intelligence.

- Built end-to-end data processing pipelines using Pandas for ingestion, cleaning, and aggregation of job listing data; rendered dynamic, drill-down visualizations with Plotly for actionable insight delivery.
- Deployed a fully live Streamlit web application with multi-filter interactivity, allowing users to explore skill demand by role, industry, and region — bridging raw job data with practical upskilling decisions.

SKILLS

Languages	Python, SQL, TypeScript
Frameworks & Tools	Next.js, Tailwind CSS, Shadcn UI, Streamlit, Vercel, Git / GitHub
Libraries	NumPy, Pandas, Matplotlib, Seaborn, Plotly, scikit-learn, Zod
Machine Learning	Regression, Classification, Random Forest, Gradient Boosting, Feature Engineering, Model Evaluation
Generative AI	LLM APIs (Groq, OpenAI), Prompt Engineering, Structured JSON Pipelines, RAG, Embeddings, Vector Databases (FAISS / Chroma), Guardrail Design
Data Analytics	EDA, Data Cleaning, Statistical Analysis, Hypothesis Testing, A/B Testing, Data Storytelling
Visualization	Matplotlib, Seaborn, Plotly, Jupyter Notebook, Google Colab
Soft Skills	Critical Thinking, Analytical Reasoning, Communication, Attention to Detail, Adaptability

CERTIFICATIONS

Machine Learning Statistical Foundations Professional Certificate

LinkedIn Learning × Wolfram Research | Completed: Mar 2026 | Skills: Machine Learning · Linear Algebra · Wolfram Language

Anaconda Python for Data Science Professional Certificate

LinkedIn Learning × Anaconda | Completed: Mar 2026 | Skills: Python · Data Analysis · Statistics

Career Essentials in Generative AI

LinkedIn Learning × Microsoft | Completed: Mar 2026 | Skills: Generative AI · Microsoft Copilot · Responsible AI